

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

TEAM ID	SWTID-2026-3007
PROJECT TITLE	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Step 1: Load the Cosmetics Dataset into Tableau

First, open Tableau Desktop.

Click Connect and select Microsoft Excel, which is the file format of the cosmetics dataset.

Choose the cosmetics dataset and ensure that all columns are imported correctly. The dataset should contain the following fields:

- Product Name
- Brand
- Category
- Price
- Rank
- Combination
- Dry
- Normal
- Oily
- Sensitive

Verify that the data types are assigned correctly:

- Product Name, Brand, and Category should be String data types.

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

- Price and Rank should be Numeric data types.
 - Combination, Dry, Normal, Oily, and Sensitive should be Numeric or Boolean data types.
-

Step 2: Clean the Data

Remove any duplicate records, if present.

Check for missing or inconsistent values and rename fields where necessary to improve readability.

Validate fields such as Price and Rank to ensure data accuracy.

After completing the cleaning process, save the cleaned data source.

Step 3: Create Filters for the Dashboard

Create a Category Filter.

1. Drag Category to the Filters shelf.
2. Select all categories and click OK.
3. Right-click the Category filter and select Show Filter.
4. Change the filter type to either Single Value Dropdown or Multiple Value Dropdown.

This filter allows users to dynamically select categories for focused analysis of the cosmetics dataset.

Step 4: Create a Calculated Field for All Skin Types Count

Create a new calculated field by navigating to Analysis → Create Calculated Field.

Name the field ALL SKIN TYPES COUNT and use the following formula:

$SUM([Combination]) + SUM([Dry]) + SUM([Normal]) + SUM([Sensitive])$

This calculated field determines the number of products suitable for multiple skin types and helps analyze product availability across the cosmetics dataset.

Step 5: Analyze Product Distribution Across Categories

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Create a new worksheet.

1. Drag Category to the Rows shelf.
2. Drag Product Name to the Columns shelf.
3. Change the aggregation to COUNT(Product Name).
4. Select a Bar Chart visualization.

This analysis displays the distribution of products across different cosmetic categories.

Step 6: Calculate Category-Wise Total Revenue

If revenue data is not directly available, use the following calculation:

$\text{SUM}([\text{Price}])$

To create the visualization:

1. Drag Category to the Rows shelf.
2. Drag Price to the Columns shelf.
3. Set the aggregation to SUM.
4. Sort the results in descending order.
5. Add data labels.

This analysis shows the revenue contribution of each category in the cosmetics dataset.

Step 7: Find the Top 10 Brands for All Skin Types

Create a calculated field using:

$[\text{Combination}] + [\text{Dry}] + [\text{Normal}] + [\text{Sensitive}]$

Then:

1. Drag Brand to the Rows shelf.
2. Drag the calculated field to the Columns shelf.
3. Sort the results in descending order.
4. Apply a Top 10 Filter to display the leading brands.

This analysis identifies brands that offer products suitable for multiple skin types.

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Step 8: Find the Top 10 Brands by Product Count

1. Drag Brand to the Rows shelf.
2. Drag Product Name to the Columns shelf.
3. Select COUNT(Product Name).
4. Sort the results in descending order.
5. Filter the view to show the top 10 brands.

This analysis identifies brands with the highest number of products.

Step 9: Compare Average Prices Across Brands

1. Drag Brand to the Rows shelf.
2. Drag Price to the Columns shelf.
3. Change the aggregation to AVG(Price).
4. Sort the results in descending order.
5. Add data labels.

This analysis compares average product prices across brands.

Step 10: Analyze the Category-Wise Distribution of Combination Skin Products

1. Drag Category to the Rows shelf.
2. Drag Combination to the Columns shelf.
3. Use SUM(Combination) as the aggregation.
4. Select a Bar Chart.

This analysis shows the availability of products designed for combination skin across categories.

Step 11: Identify Top Brands and Cosmetic Counts

1. Drag Brand to the Rows shelf.
2. Drag Product Name to the Columns shelf.
3. Use COUNT(Product Name).

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

4. Sort the results in descending order.
5. Display the top N brands.

This analysis highlights leading brands based on product volume.

Step 12: Analyze Category-Level Product Distribution

1. Drag Category to the Rows shelf.
2. Drag Product Name to the Columns shelf.
3. Use COUNT(Product Name) as the aggregation.
4. Apply color encoding based on product count.
5. Add data labels.

This analysis provides a comparative view of product distribution among categories.

Step 13: Build the Dashboard

Click New Dashboard and define the dashboard size.

Add the following worksheets:

- Category Product Distribution
- Category Revenue
- Top 10 Brands (All Skin Types)
- Top 10 Brands by Product Count
- Average Brand Price
- Combination Skin Distribution
- Top Brands and Cosmetic Counts

Arrange the visualizations logically to create a clear and user-friendly dashboard layout.

Step 14: Add Interactive Filters

1. Click the filter dropdown menu and select Apply to Worksheets.
2. Add filters for Category, Brand, and Skin Type.
3. Enable dashboard actions to allow cross-filtering and interaction.

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

As a result, all visualizations update dynamically based on user selections within the cosmetics dataset.

Step 15: Validate Dashboard Performance

Test all filter interactions and verify dashboard response times.

Check dashboard rendering and confirm that all visualizations update correctly.

Performance Summary

- Dataset Size: 1,472 rows and 10 columns
- Data Volume: Approximately 1–2 MB
- Tableau efficiently handles filtering and aggregation.
- Visualizations render smoothly with minimal response time.
- Interactive filters provide a seamless and effective data exploration experience.

